Last Updated: 4/16/2026 10:53:07 AM


PROOFS ON THE REAL NUMBERS ..... 2

Cardinality of ℝ the Real Numbers ..... 2

Prove $\exists 2^{\aleph}\ \forall 2^{\aleph}$. ..... 2

Prove $2^{n+1} \geq 2^{n}\ \forall n \in \mathbb{N}$. ..... 2

Prove $2^{n} > 0\ \forall n \in \mathbb{N}$. ..... 2

Prove $2^{n} \neq 0\ \forall n \in \mathbb{N}$. ..... 2

Prove $2^{\aleph} > 0$. ..... 2

Prove $2^{\aleph} \neq 0$. ..... 2

Prove that ℝ is a container with $\exists \mathrm{card}(\mathbb{R}) = 2^{\aleph}$. ..... 2

Prove $\exists \mathbb{R} \forall \mathbb{R}$. ..... 3

Prove ℝ is not ∅ ∀ℝ. ..... 3

The Largest Real Number is not the Largest. ..... 3

The Smallest Real Number is not the Smallest. ..... 3

The Smallest Real Granularity is not the Smallest. ..... 3

A Discussion on Extended Real Line ..... 3

有關於實數的證明 ..... 4

實數容器 ℝ 的基數 ..... 5

證明$\exists 2^{\aleph} \forall 2^{\aleph}$。 ..... 5

證明 $2^{\aleph}$即非$\varnothing \forall 2^{\aleph}$。 ..... 5

證明 $2^{n+1} \geq 2^{n}\ \forall n \in \mathbb{N}$。 ..... 5

證明 $2^{n} > 0\ \forall n \in \mathbb{N}$。 ..... 5

證明 $2^{n} \neq 0\ \forall n \in \mathbb{N}$。 ..... 5

證明 $2^{\aleph} > 0$。 ..... 5

證明 $2^{\aleph} \neq 0$。 ..... 5

證明 ℝ 即一個容器帶著$\exists \mathrm{card}(\mathbb{R}) = 2^{\aleph}$。 ..... 5

證明$\exists \mathbb{R} \forall \mathbb{R}$。 ..... 5

證明 ℝ 即非$\varnothing \forall \mathbb{R}$。 ..... 6

最大實數不是最大。 ..... 6

最小實數不是最小。 ..... 6

最細實數不是最細。 ..... 6

有關於延伸實數線的討論 ..... 6

對於ℝ的補充評論 ..... 7


All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

# PROOFS ON THE REAL NUMBERS

This section uses Huo Jian Hua's Definition of Boundedness to prove $\exists 2^{\aleph}\ \forall 2^{\aleph}$, the cardinality of $\mathbb{R}$, and then gives examples that the largest real number is not the largest, the smallest real number is not the smallest, and then shows that the smallest real granularity is not the smallest, and then briefly discusses the extended real line.

## Cardinality of $\mathbb{R}$ the Real Numbers

Write a real number in the smallest digit base as possible. The smallest digit base is 2. The sign can either be plus or minus, so the sign digit is also base 2. The base 2 representation of a real number can be in the following form:

$$b_1 b_2 b_3 \ldots b_i . b_{i+1} \ldots b_{\aleph-1} b_{\aleph}$$

where "." is the floating-point between digit $b_i$ and digit $b_{i+1}$, and $b_1$ is the sign digit. For a fixed floating-point position, there are $2^{\aleph}$ permutations. The floating-point floats from between the $b_1$ and the $b_2$ to the after of $b_{\aleph}$, so there are $\aleph 2^{\aleph}$ permutations with the floating-point. $\aleph = 2^{\frac{ln\aleph}{ln2}} \therefore \aleph 2^{\aleph} = 2^{\frac{ln\aleph}{ln}} 2^{\aleph} = 2^{\frac{ln\aleph}{ln}+\aleph} = 2^{\aleph} = \mathfrak{c}$ which is the cardinality of $\mathbb{R}$. The real number container $\mathbb{R}$ can alternatively be expressed in the interval form $\mathbb{R}=[R_1, R_c]$, where $R_1 = \min(\mathbb{R})$, and $R_c = \max(\mathbb{R})$. Note that $\mathbb{R}$ is a shorthand for $\mathbb{R}^1$.

### Prove $\exists 2^{\aleph}\ \forall 2^{\aleph}$.

Proof: Suppose $2^{\aleph}$ is infinity $\therefore$ infinity is a container. $\therefore$ infinity belongs to a container $\therefore$ infinity is bounded, by the Definition of Boundedness of The Honorable Grandmaster Ji-Gong. But, infinity is not bounded; (infinity is not bounded) and (infinity is bounded) ➔🡐 $\forall 2^{\aleph}$ $\therefore 2^{\aleph}$ is not infinity $\therefore \exists 2^{\aleph} \forall 2^{\aleph}$. Q.E.D.

Prove $2^{\aleph}$ is not $\varnothing$ $\forall 2^{\aleph}$.

Proof:

Let $2^1$, $1 \in \mathbb{N}$. There is 1 count of $2^1$ in $2^1$ $\therefore \text{card}(2^1)=1 \neq 0$ $\therefore 2^1$ is not $\varnothing$. Assume $2^n$ is not $\varnothing$, $n \in \mathbb{N}$. $2^{n+1} = (2^n)(2^1)$. Both $2^n$, and, $2^1$ are not $\varnothing$, $(2^n)(2^1)=2^{n+1}$, $(n+1) \in \mathbb{N}$. There is 1 count of $2^{n+1}$ in $2^{n+1}$ $\therefore \text{card}(2^{n+1})=1 \neq 0$ $\therefore 2^{n+1}$ is not $\varnothing$. $\therefore 2^n$ is not $\varnothing$ $\forall n \in \mathbb{N}$. $\aleph \in \mathbb{N}$ $\therefore 2^{\aleph}$ is not $\varnothing$ $\forall 2^{\aleph}$. Q.E.D.

Therefore, the existence of $2^{\aleph}$ is nontrivial.

### Prove $2^{n+1} \geq 2^n$ $\forall n \in \mathbb{N}$.

Proof: Let n=1, so n+1=2. $\therefore 2^{n+1}=2^2=4 \geq 2^n=2^1=2$. Assume $2^{n+1} \geq 2^n$.
But, $2^{(n+1)+1} \geq 2^{n+1} \Rightarrow 2(2^{(n+1)}) > 2(2^n) \Rightarrow 2(2^{(n+1)})/2 > 2(2^n)/2 \Rightarrow 2^{n+1} \geq 2^n$ $\therefore 2^{n+1} \geq 2^n$ $\forall n \in \mathbb{N}$. Q.E.D.

### Prove $2^n > 0$ $\forall n \in \mathbb{N}$.

Proof: Let n=1, $1 \in \mathbb{N}$. $2^1=2>0$. Assume $2^n>0$. But, $2^{n+1} \geq 2^n > 0$ $\therefore 2^{n+1}>0$ $\therefore 2^n>0$ $\forall n \in \mathbb{N}$. Q.E.D.

### Prove $2^n \neq 0$ $\forall n \in \mathbb{N}$.

Proof: Let n=1, $1 \in \mathbb{N}$. $2^1=2>0$, $\therefore 2 \neq 0$. Assume $2^n \neq 0$. But, $2^{n+1} \geq 2^n > 0$ $\therefore 2^{n+1}>0$ $\therefore 2^{n+1} \neq 0$. $\therefore 2^n \neq 0$ $\forall n \in \mathbb{N}$. Q.E.D.

### Prove $2^{\aleph}>0$.

Proof: $2^n>0$ $\forall n \in \mathbb{N}$. But, $\aleph \in \mathbb{N}$ $\therefore 2^{\aleph}>0$. Q.E.D.

### Prove $2^{\aleph} \neq 0$.

Proof: $2^n \neq 0$ $\forall n \in \mathbb{N}$. But, $\aleph \in \mathbb{N}$ $\therefore 2^{\aleph} \neq 0$. Q.E.D.

### Prove that $\mathbb{R}$ is a container with $\exists \text{card}(\mathbb{R})=2^{\aleph}$.

Proof: $R_1$ in $\mathbb{R}$ $\therefore \mathbb{R}$ is a container. Suppose $\text{card}(\mathbb{R})= 2^{\aleph}$ is infinity $\therefore$ infinity is bounded. But, infinity is not bounded; (infinity is not bounded) and (infinity is bounded) ➔🡐 $\forall$ $\text{card}(\mathbb{R})= 2^{\aleph}$ $\therefore$ $\text{card}(\mathbb{R})= 2^{\aleph}$ is not infinity $\therefore \exists \text{card}(\mathbb{R})= 2^{\aleph}$. Q.E.D.

All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

Prove ∃ℝ∀ℝ.

Proof: Suppose ℝ is infinity ∴ infinity is a container. But, infinity is not a container; (infinity is not a container) and (infinity is a container) ➔🡐 ∀ℝ ∴ℝ is not infinity ∴ ∃ℝ ∀ℝ. Q.E.D.

Prove ℝ is not ∅ ∀ℝ.

Proof: Suppose ℝ is ∅ ∴ card(ℝ)=0. But, card(ℝ)= $2^{\aleph}$≠0; card(ℝ)≠0 and card(ℝ)=0 ➔🡐∀ℝ ∴ℝ is not ∅ ∀ℝ. Q.E.D.

Therefore, the existence of ℝ is nontrivial. All rights-benefits of ℝ belong to Huo Jian Hua by Huo Jian Hua's contract guaranteed to Huo Jian Hua.

## The Largest Real Number is not the Largest.

The largest real number in ℝ is Max(ℝ) = $R_c$. However, $R_c$ is not the largest, for example +∞>$R_c$.

## The Smallest Real Number is not the Smallest.

The smallest real number in ℝ is Min(ℝ) = $R_1$. However, $R_1$ is not the smallest, for example -∞<$R_1$.

## The Smallest Real Granularity is not the Smallest.

Note that the smallest real granularity refers to the smallest pure real granularity. Let nonempty x∈ℝ.

| col / row | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| | Granularity | Interval v⊆ℝ | Measure u(v) | Card(v) |
| 1 | A | [-10,10] | 20 | $2^{\aleph}$ |
| 2 | B | [-π,π] | 2π | $2^{\aleph}$ |
| 3 | H | [x,x] | 0 | 1 |
| 4 | J | (x,x) | 0 | 0 |

[Table G]

From [Table G] row 3, the nonempty h has the smallest real granularity with a measure u([x,x])=0 and a cardinality card([x,x])=1. From [Table G] row 4, another granularity j is a measure u((x,x))=0 with card((x,x))=0. The granularity j and granularity h both have the same measure u(v)=0, yet the cardinality for j is smaller than the cardinality for h, so the smallest real granularity h is not the smallest because j is smaller than h by cardinality.

## A Discussion on Extended Real Line

The extended real line has been denoted by $\overline{\mathbb{R}} = [-\infty, +\infty]$. Because infinity is not a set element, in order to avoid misidentifying -∞ or +∞ being infinity while -∞ in $\overline{\mathbb{R}}$, and, while +∞ in $\overline{\mathbb{R}}$, we need to show that neither +∞ nor -∞ is infinity, and, neither +∞ nor -∞ is unbounded to avoid paradoxes of -∞ or +∞ being infinity. Also, we need to show that both -∞, and, +∞ are bounded and exist.

Prove that +∞ is not infinity ∀+∞.

Proof: Suppose +∞ is infinity ∴ +∞ in infinity ∴infinity is a container C ∴infinity belongs to C ∴infinity is bounded, by Huo Jian Hua's Definition of Boundedness. But, infinity is not bounded, (infinity is not bounded) and (infinity is bounded) ➔🡐 ∀+∞ ∴+∞ is not infinity ∀+∞. Q.E.D.

Prove that -∞ is not infinity ∀-∞.

Proof: Suppose -∞ is infinity ∴ -∞ in infinity ∴infinity is a container C ∴infinity belongs to C ∴infinity is bounded, by Huo Jian Hua's Definition of Boundedness. But, infinity is not bounded, (infinity is not bounded) and (infinity is bounded) ➔🡐 ∀-∞ ∴-∞ is not infinity ∀-∞. Q.E.D.

All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

Show that $+\infty$ is bounded $\forall+\infty$.
Proof: $0<+\infty$, $0\in\overline{\mathbb{R}}$, so $+\infty$ is bounded from below by 0 $\therefore+\infty$ is bounded $\forall+\infty$. ■

Because of the "infinity is unbounded" axiom, and, because of $+\infty=\infty$ has been assumed in practice, we have removed "$+\infty$ is not unbounded" proof, so the axiom holds.

Show that $-\infty$ is bounded $\forall-\infty$.
Proof: $0>-\infty$, $0\in\overline{\mathbb{R}}$, so $-\infty$ is bounded from above by 0 $\therefore-\infty$ is bounded $\forall-\infty$. ■



Prove that $+\infty$ is bounded $\forall+\infty\in\overline{\mathbb{R}}$.
Proof: $+\infty\in\overline{\mathbb{R}}$ $\therefore+\infty$ in $\overline{\mathbb{R}}$ $\therefore$ $+\infty$ belongs to $\overline{\mathbb{R}}$ $\therefore$by Huo Jian Hua's Definition of Boundedness, $+\infty$ is bounded $\forall+\infty$. Q.E.D.

Prove that $-\infty$ is bounded $\forall-\infty\in\overline{\mathbb{R}}$.
Proof: $-\infty\in\overline{\mathbb{R}}$ $\therefore-\infty$ in $\overline{\mathbb{R}}$ $\therefore$ $-\infty$ belongs to $\overline{\mathbb{R}}$ $\therefore$by Huo Jian Hua's Definition of Boundedness, $-\infty$ is bounded $\forall-\infty$. Q.E.D.

Further, both $-\infty$, and, $+\infty$ are elements.

Prove $-\infty$ is an element $\forall-\infty$.
Proof: $-\infty$ in $\overline{\mathbb{R}}$ $\therefore-\infty\in\overline{\mathbb{R}}$ $\therefore-\infty$ is an element of $\overline{\mathbb{R}}$ $\therefore$ $-\infty$ is an element $\forall-\infty$. Q.E.D.

Prove $+\infty$ is an element $\forall+\infty$.
Proof: $+\infty$ in $\overline{\mathbb{R}}$ $\therefore+\infty\in\overline{\mathbb{R}}$ $\therefore+\infty$ is an element of $\overline{\mathbb{R}}$ $\therefore$ $+\infty$ is an element $\forall+\infty$. Q.E.D.

Prove $\exists-\infty\forall-\infty\in\overline{\mathbb{R}}$.
Proof: $\because-\infty$ is not infinity $\therefore\exists-\infty\forall-\infty\in\overline{\mathbb{R}}$.

Prove $\exists+\infty\forall+\infty\in\overline{\mathbb{R}}$.
Proof: $\because+\infty$ is not infinity $\therefore\exists+\infty\forall+\infty\in\overline{\mathbb{R}}$.

While $+\infty$, and, $-\infty$ are elements in $\overline{\mathbb{R}}=[-\infty,+\infty]$, neither $+\infty$ nor $-\infty$ is infinity; neither $+\infty$ nor $-\infty$ is unbounded. Furthermore, both $-\infty$, and, $+\infty$ are bounded and exist. So therefore, by Huo Jian Hua's Definition of Boundedness contract guaranteed to Huo Jian Hua, all rights-benefits of $\overline{\mathbb{R}}$ go to Huo Jian Hua, and all rest of rights-benefits go to Huo Jian Hua because "Grandmaster's guaranteed us to prove that infinity is not infinity" has been honored.

A Comment on $\overline{\mathbb{R}}$

$\overline{\overline{\mathbb{R}}}=\mathbb{R}\cup\{-\infty,+\infty\}$ has been used to denote $\overline{\mathbb{R}}$, so $\overline{\overline{\mathbb{R}_o}}=\{-\infty\}\oplus\mathbb{R}\oplus\{+\infty\}$ is an ordered set for $\overline{\overline{\mathbb{R}}}$, but neither $\overline{\overline{\mathbb{R}}}$ nor $\overline{\overline{\mathbb{R}_o}}$ captures $(-\infty, R_1)$ or $(R_c, +\infty)$ while $(-\infty, R_1)\subseteq[-\infty,+\infty]$ or $(R_c, +\infty)\subseteq[-\infty,+\infty]$. Because $\overline{\overline{\mathbb{R}}}$ or $\overline{\overline{\mathbb{R}_o}}$ might not have elements that $\overline{\mathbb{R}}$ has, so $\overline{\overline{\mathbb{R}}}$ or $\overline{\overline{\mathbb{R}_o}}$ might not be a sufficient expression for $\overline{\mathbb{R}}$.

We have shown applications with the use of Huo Jian Hua's Definition of Boundedness, to prove $\exists2^{\aleph}\ \forall2^{\aleph}$. Then we have defined $\mathbb{R}=[R_1, R_c]$. The largest real number $R_c$ is not the largest because $+\infty$ is larger than $R_c$. The smallest real number $R_1$ is not the smallest because $-\infty$ smaller than $R_1$. Further, the smallest real granularity [x,x] is not the smallest because (x,x) is smaller than [x,x] by cardinality. We have also discussed the extended real numbers $\overline{\mathbb{R}}=[-\infty,+\infty]$. All rights-benefits go to Huo Jian Hua, and all rest of rights-benefits go to Huo Jian Hua by Huo Jian Hua's Definition of Boundedness contract guaranteed to Huo Jian Hua.

## 有關於實數的證明

此篇文章使用霍建華〈有界的定義〉為 $\mathbb{R}$ 的基數證明$\exists2^{\aleph}\ \forall2^{\aleph}$。然後此文給予最大實數不是最大的舉例，最小實數不是最小的舉例，最細實數不是最細的舉例，以及有關延伸實數線的討論。

All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

## 實數容器 ℝ 的基數

2 是最小的底數可用以寫下任何一個實數。正負符號可為正或負，所以正負符號也依 2 為底數。以下是可依 2 底數表達格式的實數：

$$b_1b_2b_3\ldots b_i.\,b_{i+1}\ldots b_{\aleph-1}b_{\aleph}$$

而「. 」是小數點介於數位$b_i$與數位$b_{i+1}$之間以及$b_1$為正負符號數位。在固定小數點位置的狀況下，則有 $2^{\aleph}$ 個的排列。小數點可從$b_1$與$b_2$之間浮動至$b_{\aleph}$之後，因此其有$\aleph 2^{\aleph}$ 排列在有浮動小數點的情況下。$\aleph = 2^{\frac{ln\aleph}{ln2}}$ ∴ $\aleph 2^{\aleph} = 2^{\frac{ln\aleph}{ln2}}2^{\aleph} = 2^{\frac{ln\aleph}{ln}+\aleph} = 2^{\aleph} = \mathfrak{c}$ 為 ℝ 的基數。可以另以間隔表達格式來表達實數容器 ℝ： ℝ=[$R_1$,$R_c$]，$R_1$ =最小（ℝ）， 以及$R_c$=最大（ℝ）。備註：ℝ 是$\mathbb{R}^1$的速寫。



### 證明$\exists 2^{\aleph}\forall 2^{\aleph}$。

證： 假設$2^{\aleph}$即無限∴無限即一個容器 ∴無限屬於一個容器 ∴無限即有界，根據濟公師尊的〈有界的定義〉。但，無限即非有界，（無限即非有界）又（無限即有界）➔🡐 $\forall 2^{\aleph}$ ∴$2^{\aleph}$即非無限 ∴ $\exists 2^{\aleph}\forall 2^{\aleph}$。證明完畢。

### 證明 $2^{\aleph}$即非$\varnothing\forall 2^{\aleph}$。

證：

設 $2^1$，1∈ℕ。有 1 個 $2^1$ 在 $2^1$ 內 ∴card($2^1$)=1≠0 ∴$2^1$ 即非∅。認定 $2^n$ 即非∅，n∈ℕ。
$2^{n+1}$ = ($2^n$)($2^1$)。$2^n$ 及 $2^1$ 皆非∅，∴($2^n$)($2^1$)=$2^{n+1}$，(n+1)，（n+1）∈ℕ。有 1 個 $2^{n+1}$ 在 $2^{n+1}$ 內∴card($2^{n+1}$)=1≠0 ∴$2^{n+1}$ 即非∅。∴$2^n$ 即非∅∀n∈ℕ。ℵ∈ℕ ∴$2^{\aleph}$ 即非∅ $\forall 2^{\aleph}$。證明完畢。

所以，$2^{\aleph}$ 的存在並非微不足道。

### 證明 $2^{n+1}\geq 2^n$ ∀n∈ℕ。

證： 設 n=1，所以 n+1=2 ∴ $2^{n+1}=2^2=4\geq 2^n=2^1=2$。認為 $2^{n+1}\geq 2^n$。
但，$2^{(n+1)+1}\geq 2^{n+1}$ ⇒ $2(2^{(n+1)})>2(2^n)$ ⇒ $2(2^{(n+1)})/2 >2(2^n)/2$ ⇒ $2^{n+1}\geq 2^n$ ∴$2^{n+1}\geq 2^n$ ∀n∈ℕ。證明完畢。

### 證明 $2^n>0$ ∀n∈ℕ。

證明： 設 n=1，1∈ℕ。$2^1=2>0$。認為 $2^n>0$。但，$2^{n+1}\geq 2^n>0$∴$2^n>0$ ∀n∈ℕ。證明完畢。

### 證明 $2^n\neq 0$ ∀n∈ℕ。

證明： 設 n=1，1∈ℕ。$2^1=2>0$ ∴2≠0。認為 $2^n\neq 0$。但，$2^{n+1}\geq 2^n>0$ ∴$2^{n+1}>0$ ∴$2^{n+1}\neq 0$ ∴$2^n\neq 0$ ∀n∈ℕ。證明完畢。

### 證明 $2^{\aleph}>0$。

證： $2^n>0$ ∀n∈ℕ。但，ℵ∈ℕ∴$2^{\aleph}>0$。證明完畢。

### 證明 $2^{\aleph}\neq 0$。

證： $2^n\neq 0$ ∀n∈ℕ。但，ℵ∈ℕ∴$2^{\aleph}\neq 0$。證明完畢。

### 證明 ℝ 即一個容器帶著∃card(ℝ)=$2^{\aleph}$。

證： $R_1$在 ℝ 內 ∴ℝ 即一個容器。假設 card(ℝ)= $2^{\aleph}$即無限 ∴無限即有界。但，無限即非有界，（無限即非有界）又（無限即有界）➔🡐 ∀ card(ℝ)= $2^{\aleph}$ ∴ card(ℝ)= $2^{\aleph}$即非無限 ∴ ∃card(ℝ)= $2^{\aleph}$。證明完畢。

### 證明∃ℝ∀ℝ。

證： 假設 ℝ 即無限 ∴無限即一個容器。但，無限即非一個容器，（無限即非一個容器）又（無限即一個容器）➔🡐 ∀ℝ ∴ℝ 即非無限∴ ∃ℝ ∀ℝ。證明完畢。

All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

證明 ℝ 即非∅∀ℝ。

證： 假設 ℝ 即∅ ∴ card(ℝ)=0。但，card(ℝ)=$2^{\aleph}$≠0。card(ℝ)≠0 又 card(ℝ)=0 ➔🡐∀ℝ ∴ℝ 即非∅∀ℝ。證明完畢。

所以，ℝ 的存在並非微不足道。一切 ℝ 的權力益全都歸霍建華，根據霍建華〈有界的定義〉使用契約保證。

## 最大實數不是最大。

在 ℝ 內最大的實數即$R_c$。然而$R_c$不是最大，例如：$+\infty>R_c$。

## 最小實數不是最小。

在 ℝ 內最小的實數即$R_1$。然而$R_1$不是最小，例如：$-\infty<R_1$。

## 最細實數不是最細。

注意最細實數是指最細的純實數。設非空無 x∈ℝ。

| 排 / 行 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| | 細度 | 間隔 v⊆ℝ | 量度 u(v) | 基數 Card(v) |
| 1 | a | [-10, 10] | 20 | $2^{\aleph}$ |
| 2 | b | [-π, π] | 2π | $2^{\aleph}$ |
| 3 | h | [x, x] | 0 | 1 |
| 4 | j | (x, x) | 0 | 0 |

[覽 G]

從[覽 G]行 3 裡面的非空無 h 擁有最細實數其量度 u([x, x])=0 及基數 card([x, x])=1。另外，從[覽 G]行 4 裡有細度 j，其量度為 u((x, x))=0 帶著基數 card((x, x))=0。細度 J 及細度 h 皆有著相同的量度 u(v)=0，然而 j 所賦的基數小於 h 所賦的基數，因此最細的實數 h 不是最細，因為 j 以基數小於 h。

## 有關於延伸實數線的討論

延伸實數線通常以$\overline{\mathbb{R}}=[-\infty,+\infty]$做為表示。由於無限不是一個集合元素，為了避免誤認為$-\infty$或$+\infty$就是無限，而$-\infty$及$+\infty$又都在$\overline{\mathbb{R}}$之內，我們需要說明$+\infty$及$-\infty$皆非無限，而且 $-\infty$及$+\infty$皆非無界，以避免$-\infty$或 $+\infty$被誤當作無限所引起的悖論。此外，我們還需要說明$-\infty$及$+\infty$皆有界且皆存在。

證明$+\infty$即非無限∀$+\infty$。

證： 假設$+\infty$即無限，則$+\infty$在無限內∴無限即一容器 C，∴無限屬於 C，∴無限即有界，依據霍建華的〈有界的定義〉。但是，無限即非有界；（無限即非有界）又（無限即有界）➔🡐∀$+\infty$，∴$+\infty$即非無限。證明完畢。

證明$-\infty$即非無限∀$-\infty$。

證： 假設$-\infty$即無限，則 $-\infty$在無限內 ∴ 無限即一容器 C，∴無限屬於 C，∴無限即有界，依據霍建華的〈有界的定義〉。但是，無限即非有界；（無限即非有界）又（無限即有界）➔🡐∀$-\infty$，∴$-\infty$即非無限。證明完畢。

由於「無限即無界」公理，又，因為$+\infty=\infty$是已被採用的慣常做法，我們已放棄「$+\infty$即非無界」的證明以公理支承。



All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.

展示$+\infty$是有界的 $\forall+\infty$。
證： $0<+\infty$， 而 $0\in\overline{\mathbb{R}}$，所以 0 為$+\infty$的下界，∴ $+\infty$是有界的$\forall+\infty$。■

展示$-\infty$是有界的 $\forall-\infty$。
證： $0>-\infty$， 而 $0\in\overline{\mathbb{R}}$，所以 0 為$+\infty$的上界，∴ $-\infty$是有界的$\forall-\infty$。■

證明$+\infty$即有界$\forall+\infty\in\overline{\mathbb{R}}$。
證: $+\infty\in\overline{\mathbb{R}}$ ∴$+\infty$在$\overline{\mathbb{R}}$內∴$+\infty$屬於$\overline{\mathbb{R}}$ ∴根據霍建華〈有界的定義〉$+\infty$即有界$\forall+\infty$。證明完畢。

證明$-\infty$即有界$\forall-\infty\in\overline{\mathbb{R}}$。
證: $-\infty\in\overline{\mathbb{R}}$ ∴$-\infty$在$\overline{\mathbb{R}}$內∴$-\infty$屬於$\overline{\mathbb{R}}$ ∴根據霍建華〈有界的定義〉$-\infty$即有界$\forall-\infty$。證明完畢。

此外，$-\infty$ 及 $+\infty$ 皆為元素。

證明 $-\infty$ 即元素 $\forall-\infty$。
證： $-\infty$在$\overline{\mathbb{R}}$內 ∴$-\infty\in\overline{\mathbb{R}}$ ∴ $-\infty$即$\overline{\mathbb{R}}$的元素，∴ $-\infty$即元素$\forall-\infty$。證明完畢。

證明 $+\infty$ 即元素 $\forall+\infty$。
證： $+\infty$在$\overline{\mathbb{R}}$內 ∴$+\infty\in\overline{\mathbb{R}}$ ∴ $+\infty$即$\overline{\mathbb{R}}$的元素，∴ $+\infty$即元素$\forall+\infty$。證明完畢。

證明 $\exists-\infty\forall-\infty\in\overline{\mathbb{R}}$。
證： ∵$-\infty$即非無限，∴$\exists-\infty\forall-\infty\in\overline{\mathbb{R}}$。

證明 $\exists+\infty\forall+\infty\in\overline{\mathbb{R}}$。
證： ∵$+\infty$即非無限，∴$\exists+\infty\forall+\infty\in\overline{\mathbb{R}}$。

在$+\infty$及$-\infty$皆為是$\overline{\mathbb{R}}=[-\infty,+\infty]$的元素同時， $+\infty$及$-\infty$皆非無限，$+\infty$及$-\infty$也皆非無界。此外，$-\infty$ 及$+\infty$皆有界且存在的。因此，所以根據對霍建華保證的霍建華〈有界的定義〉使用契約，全部$\overline{\mathbb{R}}$的全部權力益全都歸霍建華，因為濟公保證要證明無限不是無限 — 兌現了。

## 對於$\overline{\mathbb{R}}$的補充評論

$\overline{\overline{\mathbb{R}}}=\mathbb{R}\cup\{-\infty,+\infty\}$已被用來表示$\overline{\mathbb{R}}$，所以$\overline{\overline{\mathbb{R}_o}}=\{-\infty\}\oplus\mathbb{R}\oplus\{+\infty\}$是$\overline{\overline{\mathbb{R}}}$的一個有序集合。然而，$\overline{\overline{\mathbb{R}}}$及$\overline{\overline{\mathbb{R}_o}}$皆並不能捕捉到$(-\infty,R_1)$或 $(R_c,+\infty)$，雖然$(-\infty,R_1)\subseteq[-\infty,+\infty]$或$(R_c,+\infty)\subseteq[-\infty,+\infty]$。因為$\overline{\overline{\mathbb{R}}}$或$\overline{\overline{\mathbb{R}_o}}$可能沒有$\overline{\mathbb{R}}$所有的元素，所以$\overline{\overline{\mathbb{R}}}$或$\overline{\overline{\mathbb{R}_o}}$可能不足以完整表達$\overline{\mathbb{R}}$。

我們經證明$\exists2^{\aleph}\forall2^{\aleph}$展示了霍建華〈有界的定義〉的應用。然後我們定義了 $\mathbb{R}=[R_1,\ R_c]$以展示最大的實數$R_c$不是最大因為$+\infty$比$R_c$還更大，最小的實數$R_1$不是最小因為$-\infty$比$R_1$還更小，最細的實數$[x,x]$不是最細因為$(x,x)$比$[x,x]$的基數還更小，我們也討論了$\overline{\mathbb{R}}=[-\infty,+\infty]$。一切權力益全都歸霍建華，全部剩下的權力益全都歸霍建華，根據霍建華〈有界的定義〉使用契約保證。



All rights-benefits go to 霍建華 (Huo Jian Hua) by Hui Jian Hua's Definition of Boundedness contract guaranteed Huo Jian Hua.